

Understanding AMIRIS

A complete, plain-language reference: what goes in, what the agents do, how a price gets set, and real worked examples from our own 2027, 2028 and 2029 models

This document is a standing reference, not a log of work done. Its purpose is to let you answer almost any question about how the model itself works - what data it consumes, what each of the roughly 30 agents in the scenario actually represents in the real electricity system, how they interact to produce an hourly price, and what that looks like in practice on real hours pulled straight from the current-best build for each of the three years built so far: 2027 (Germany2027_Feb29DropFix), 2028 (Germany2028), and 2029 (Germany2029_Feb29DropFix). Every example number in Parts 5 and 8 was read directly from that build's real result files and balance-checked (supply exactly equals demand plus storage charging, for every example hour) - nothing here is illustrative or invented.

In plain terms: Think of AMIRIS as a very detailed, hour-by-hour re-enactment of Germany's real electricity exchange, played out by about 30 separate computer 'actors' - one for each power plant type, each renewable source, each storage unit, the demand side, and the import channel. Every simulated hour, all of them submit bids into an auction, exactly the way real power plant operators do on the real German exchange, and the auction produces one clearing price for that hour.

Parts 1-6 below explain the mechanics using the 2027 build as the worked example. Parts 7-9 then extend the same explanation to the 2028 and 2029 out-of-sample validation builds - what changes year to year (capacities, demand growth), what stays identical (every mechanism, every rule), and real verified examples from each year's own results.

Part 1 - The Big Picture

1.1 What AMIRIS Actually Is

AMIRIS (Agent-based Market model for the Investigation of Renewable and Integrated energy Systems) is an agent-based simulation of Germany's day-ahead electricity market, built by the DLR (German Aerospace Center). 'Agent-based' means the model doesn't calculate one big central optimization for the whole country at once - instead, it creates a separate small program ('agent') for every real market participant type: each fuel's power plant fleet, each renewable technology, each storage unit, the demand side, and cross-border imports. Every simulated hour of the year, each agent independently decides what to bid, and a market-clearing agent runs an auction across all those bids - just like the real day-ahead electricity exchange does.

We are using AMIRIS to try to reproduce Germany's real 2027 (and 2028/2029) electricity prices as closely as possible, using Energy Brainpool's real input data (capacities, fuel costs, demand, weather-driven renewable profiles) as the model's inputs, then comparing AMIRIS's own simulated price against Brainpool's own commercially-forecast price for the same hours. The whole project is that comparison.

1.2 The Three-Stage Flow

Stage	What Happens	Example
1. INPUTS	Real-world data files describing capacities, costs, weather-driven output, and demand - supplied mostly from Energy Brainpool's data, filled in with AMIRIS's own defaults only where Brainpool has a genuine gap.	load_2027_feb29fix.csv (hourly demand in MWh), wind_onshore_profile.csv (hourly wind output as a share of installed capacity)
2. AGENTS / MARKET	Every simulated hour, each agent computes a bid based on its inputs (fuel cost, weather, demand, storage state) and submits it. A single market-clearing agent (the exchange) collects every bid and clears the auction.	A gas plant bids its fuel+CO2 cost plus a markup; a wind farm bids near zero (it has almost no running cost); the exchange sorts all bids cheapest-first until total supply meets total demand
3. OUTPUTS	One cleared price per hour for the whole year (8,760 or 8,784 hours), plus how much energy every single agent was awarded that hour.	DayAheadMarketSingleZone.csv: one row per hour, with the clearing price and total traded volume; separate CSVs per agent type showing exactly what each agent did

1.3 What Happens Inside One Simulated Hour (the hidden clockwork)

The 'contracts' files (contracts/*.yaml) define a strict messaging timeline for every single simulated hour, using relative time offsets in seconds-before/after the hour actually clears. This is the part that's invisible in the output but explains WHY the model needs forecasting agents at all:

Offset	What Happens
-30	The exchange sends a 'gate closure' notice to every trader - the deadline for that hour's bidding round.
-26 to -21	The SensitivityForecaster asks every trading agent for a price forecast, and gets one back - this is how storage and flexible-demand agents (battery, electrolysis, EVs) get a sense of whether the coming hours will be cheap or expensive, so they can decide whether it's worth waiting.
-24 to -22	Every conventional plant operator asks the fuel market and carbon market for a FORECAST of fuel/CO2 prices (not the real price yet), and computes a forecast marginal cost.
-9	Traders forward the real gate-closure deadline down to the plant operators they represent.
-3 to -1	Plant operators now request the REAL (not forecast) fuel and CO2 price, compute their real marginal cost, and hand it to their trader.
0	Every trader (conventional, renewable, storage, import, demand) submits its actual bid to the exchange. This is the moment of the auction.
+4	The exchange has cleared the auction and sends back Awards - how much energy each agent actually got to sell or buy, and at what price.
+5 to +6	Traders tell their plant operators to actually dispatch that amount, and pay them.

In plain terms: This is why the model needs a forecaster agent at all: a battery deciding whether to charge THIS hour needs some idea of whether prices will be cheaper or more expensive later in the week, before it commits. The forecast pass (offsets -26 to -19) is the model's way of giving every flexible agent a look-ahead, the same way a real trading desk would check a price forecast before deciding whether to buy now or wait.

Part 2 - The Inputs: What Goes Into the Model

Every input is a plain CSV time series (one value per hour, for the whole year) or a fixed number in a scenario file. Everything below is a real file actually used in the Germany2027_Feb29DropFix build.

Category	Real File(s)	What It Contains
Demand	load_2027_feb29fix.csv	Hourly electricity demand in MWh for the whole of Germany - includes the inflexible base load, heat pumps, and (as of Phase 20/21) used to include electrolysis/e-mobility too, until those were pulled out into their own price-responsive agents (Part 3.4). ValueOfLostLoad=3000 EUR/MWh is set alongside this - the price the model charges itself if it literally cannot meet demand.
Renewable weather profiles	solar_openfield_profile.csv, solar_rooftop_profile.csv, wind_onshore_profile.csv, wind_offshore_profile.csv, run_of_river_profile.csv, biomass_profile.csv	Hour-by-hour output as a fraction of installed capacity, driven by real historical weather years (see Part 3.3) - this is what makes solar output zero at night and wind output vary with actual wind speed.
Fuel prices	hard_coal_price.csv, natural_gas_price.csv, oil_price.csv (lignite is a flat 5.00 EUR/MWh, nuclear is unused - 0 GW)	Hourly fuel cost per MWh (thermal), built from Brainpool's Variable_cost sheet, used by conventional plant operators to compute their marginal cost each hour.
CO2 price	co2_price.csv	Hourly carbon price, multiplied by each fuel's emissions factor (e.g. lignite 0.407 t/MWh) to add a carbon cost on top of fuel cost.
Plant availability	hard_coal_outage.csv, lignite_outage.csv, natural_gas_outage.csv (+ matching must_run.csv files)	Hourly factor reducing a fleet's available capacity (planned/unplanned outages) and a separate factor forcing a minimum must-run output (some plants can't fully shut off).
Import availability & cost	AvailableEnergyForImport.csv, ImportCostInEURperMWh.csv	How much power could be imported from neighboring countries each hour (capped at a 30,000 MW ceiling) and at what price - this is Germany's 'relief valve' when domestic supply is tight.
Subsidy / support rates	Set directly in RenewablesAndPolicy.yaml, not a timeseries	Fixed feed-in-tariff (FIT) or market-premium (MPVAR) rates per renewable technology - real EEG/BNetzA auction figures where available (e.g. solar openfield 49 EUR/MWh, wind onshore 52 EUR/MWh), documented placeholders where no real 2027 figure exists yet (e.g. offshore wind LCOE, since 2025's real offshore auctions received zero bids).

Part 3 - The Agents: Who's Who In the Simulation

The Germany2027_Feb29DropFix scenario defines 30 agents in total. They fall into five functional groups. Every agent Id below is the real Id used in the actual scenario files.

3.1 Market Infrastructure (the plumbing, not a producer or consumer)

Agent (Id)	Role
DayAheadMarketSingleZone (1)	The exchange itself. Collects every bid each hour, clears the auction (cheapest bids served first, all winners paid the same clearing price), and publishes the result - this IS the auction, not a participant in it.
CarbonMarket (3)	Supplies the CO2 price (real and forecast) that every fossil plant operator needs to compute its true marginal cost.
FuelsMarket (4)	Supplies each fuel's price (real and forecast) the same way.
SensitivityForecaster (6)	Gives every storage/flexible agent (battery, electrolysis, EVs, pumped hydro) a forward price estimate over the next ~730 hours, so they can plan WHEN to charge or discharge rather than reacting blindly hour to hour.

3.2 Conventional (Fossil) Generation - Three Agents Per Fuel Type

Each fossil fuel is modelled with a stack of three linked agents that mirror how a real power company is organised: an owner who builds the fleet, an operator who runs it and knows its true cost, and a trader who actually deals in the market.

Agent Role	What It Does	Real Fleets in Germany2027
PredefinedPlantBuilder	Defines the physical fleet: total installed MW, efficiency range, block size, CO2 factor. Fixed for the whole year - doesn't change hour to hour.	Lignite 13,927 MW (Id 2001) - Hard Coal 5,504 MW (Id 2002) - Natural Gas (CCGT+OCGT combined) 36,237 MW (Id 2003) - Oil+Other Fossil combined 9,790 MW (Id 2005)
ConventionalPlantOperator	Every hour, computes the real marginal cost to run (fuel cost / efficiency + CO2 price x emissions factor + variable opex), applies that hour's outage/must-run factor, and reports the true cost to its trader.	Ids 501 (lignite), 502 (hard coal), 503 (gas), 505 (oil)
ConventionalTrader	Takes the operator's marginal cost, adds a markup within a fixed band, and submits the actual bid to the exchange.	Ids 1001-1005, e.g. gas allowed -10 to +10 EUR/MWh markup, lignite -60 to 0 (lignite bids can go well below its own cost to guarantee dispatch, since shutting down and restarting is expensive)

Note on nuclear and gas splitting:

Nuclear is defined in the schema but not built (0 GW - reflects the real German nuclear phase-out). Natural gas is modelled as ONE combined CCGT+OCGT block rather than two separate ones, because Brainpool's real data gives no split between the two gas plant types - AMIRIS's own CCGT efficiency band was used as the representative choice.

3.3 Renewable Generation

Agent (Id)	Real Capacity	Weather Source
Solar Openfield - VariableRenewableOperator (60)	119,179 MW	Real historical solar irradiance year
Solar Rooftop - VariableRenewableOperator (61)	33,706 MW	Same solar weather year, rooftop-orientation-adjusted
Wind Onshore - VariableRenewableOperator (70)	90,232 MW	Real historical wind year (2016 base)
Wind Offshore - VariableRenewableOperator (80)	12,573 MW	AMIRIS's own profile, since no real 2027-specific offshore reference exists - validated against real German offshore fleet capacity-factor benchmarks (37-45%) as a sanity check
Run-of-River - VariableRenewableOperator (50)	4,161 MW	Weather/river-flow driven, modelled as a must-take renewable (not dispatchable storage, unlike Reservoir Hydro in 3.4)

Agent (Id)	Real Capacity	Weather Source
Biogas (52)	8,250 MW	NOT weather-driven - uses a fixed real dispatch shape (AMIRIS's own biomass dispatch convention), since biogas plants can be scheduled rather than depending on weather

All renewable generators bid at (or very near) zero marginal cost - the wind and sun are free once the turbine/panel is built - so they are almost always the cheapest bids in the auction and get dispatched first, ahead of every fossil plant. Two supporting agents handle the subsidy side: RenewableTrader (Id 11) collects any market revenue, and SupportPolicy (Id 90) pays the FIT or market-premium (MPVAR) top-up defined per technology.

3.4 Storage & Flexible Demand - All the Same Agent Type, Configured Differently

Every storage-like agent uses the identical underlying agent type, GenericFlexibilityTrader - what differs between them is purely the numbers in their configuration.

Agent (Id)	Power / Energy	Behaviour
Pumped Hydro (700)	8,378 MW / 53,608 MWh (6.4h duration)	Traditional two-way storage - MIN_SYSTEM_COST assessment, meaning it charges/discharges to minimise overall system cost rather than to maximise its own profit.
Battery (701)	5,152 MW / 10,303 MWh (2.0h duration)	Two-way storage - MAX_PROFIT assessment, buys low and sells high purely to make money, which in practice smooths price spikes.
Reservoir Hydro (702)	1,540 MW / 171,282 MWh (111h duration)	Two-way, but very long-duration - can shift energy across weeks, not just within a day.
Electrolysis (704)	2,500 MW charge / 100,000 MWh buffer	ONE-WAY forced consumer: a constant -1,708.45 MW drain (representing 14.966 TWh/year of mandatory hydrogen production) continuously empties its buffer, and MAX_PROFIT decides WHEN within that buffer's slack to recharge - i.e. it always consumes the same total energy over the year, but shifts WHEN, toward cheap or negative-price hours.
E-Mobility (705)	6,000 MW charge / 40,000 MWh buffer	Same one-way pattern as electrolysis: a constant -2,032.78 MW drain (17.807 TWh/year of driving demand), smart-timed recharging. Smaller buffer (0.8 days vs. electrolysis's 2.4 days) because real drivers need their cars back, unlike an industrial process.

In plain terms: Electrolysis and e-mobility are NOT free traders like the battery - they are forced to consume a fixed total amount of energy over the year no matter what (that's the constant drain). The only thing MAX_PROFIT controls is timing: like a phone that MUST be charged by morning, but is smart enough to wait for the cheapest hour of the night to actually do it, rather than starting to charge the moment it's plugged in.

3.5 Demand and Import

Agent (Id)	Role
DemandTrader (100)	Represents all inflexible German electricity demand. Bids up to ValueOfLostLoad (3,000 EUR/MWh) every hour, guaranteeing it always gets served unless supply is completely exhausted - this is what makes price spike to exactly 3,000 in a genuine shortage (see Part 5.2).
ImportTrader (112)	Offers imported electricity from neighbouring countries, up to a 30,000 MW ceiling, at a real hourly cost curve. Acts as a release valve: cheap when the ceiling isn't binding, but capped once 30,000 MW is reached, same as it would be in reality (interconnector capacity is physically limited).

Part 4 - How a Price Actually Gets Set

4.1 The Auction Mechanism

Every hour, the exchange (DayAheadMarketSingleZone) receives a bid from every trader: a quantity and a price. It sorts ALL bids from cheapest to most expensive (this ordering is the real-world 'merit order' - renewables and low-cost lignite first, expensive gas/oil peakers last) and accepts bids in that order until total accepted supply exactly equals total demand for that hour. Every accepted bid - even the very cheapest wind bid - gets paid the SAME clearing price: the price of the last (most expensive) bid needed to meet demand. This is a uniform-price auction, the same design the real German day-ahead market uses.

4.2 What Pushes Price Up: Shortage

If total available supply (all conventional plants running near their limits, all renewables and storage already dispatched, import maxed at its 30,000 MW ceiling) still cannot fully cover demand, the DemandTrader's own bid - at ValueOfLostLoad, 3,000 EUR/MWh - becomes the marginal (price-setting) bid. Price hits exactly 3,000 EUR/MWh in these hours. This happened for 2 hours in the real 2027 build (see Part 5.2).

4.3 What Pushes Price Down: Negative Prices

When renewable output alone is close to or exceeds total demand, and storage/flexible agents can't absorb all of the excess, some generators are still willing to bid below zero - effectively paying to keep running - because for MPVAR/FIT-subsidised renewables, the subsidy income can outweigh a mildly negative market price, and for some conventional plants, shutting down and restarting costs more than a few hours of negative-price operation. The market clears at whatever negative price is needed to bring supply back down to demand. This happened 1,608 hours in the real 2027 build - 18.4% of the year - concentrated April through November (see Part 6.1).

Part 5 - Five Real Hours From the Actual 2027 Results

Every number below is read directly from Germany2027_Feb29DropFix's real result files (DayAheadMarketSingleZone.csv, DemandTrader.csv, VariableRenewableOperator.csv, Biogas.csv, ConventionalPlantOperator.csv, GenericFlexibilityTrader.csv, ImportTrader.csv) and balance-checked: total supply (renewables + biogas + conventional + import + storage discharge) exactly equals total consumption (demand + storage/flex charging) for every hour shown.

5.1 Ordinary Winter Evening Peak - 19 January, 18:00

Timestamp: 19 Jan 2027, 18:00 (a cold winter evening - everyone home, lights and heating on) Clearing price: 75.89 EUR/MWh (moderate)

Supply source	MWh
Demand	99,261
Wind onshore	70,000 (near its full ~90 GW capacity - a very windy evening)
Solar (openfield + rooftop)	0 (dark - 18:00 in January)
Wind offshore + run-of-river	5,205
Biogas	5,244
Conventional (gas 15,000 + lignite 8,376 + coal 1,624)	25,001
Import	0 (not needed)
Storage/flex charging (topping up while still affordable)	3,780

This is the year's single highest-demand hour, yet the price stayed moderate - because a very windy evening meant onshore wind alone covered 70,000 MWh of the near-100,000 MWh demand, so only a modest amount of conventional generation was needed to fill the rest. No import was required at all. This is a good illustration that peak DEMAND and peak PRICE are not the same thing - what matters is demand relative to available supply that hour.

5.2 Shortage Hour - the Model Hits Its Price Ceiling - 17 December, 06:00

Timestamp: 17 Dec 2027, 06:00 (a calm, dark December morning) Clearing price: 3,000.00 EUR/MWh (the ValueOfLostLoad ceiling - a genuine shortage signal)

Supply source	MWh
Demand	85,237
Wind + solar + hydro (calm, dark morning - very low)	4,687
Biogas	5,190
Conventional (running near its practical maximum)	49,017
Import (maxed at the full 30,000 MW ceiling)	30,000
Storage discharge (drawing down reserves)	1,784
Storage/flex still charging a little elsewhere	5,441

Even with conventional plants running hard, import maxed out at its full 30,000 MW ceiling, and storage discharging reserves, total available supply still could not comfortably clear demand - so the DemandTrader's own 3,000 EUR/MWh bid became the marginal price. This is exactly what ValueOfLostLoad is designed to do: signal 'the model has run out of everything it can throw at this hour, including imports.' It happened for only 2 hours in the whole real 2027 build (06:00 and 07:00 that same morning) - a rare, genuine extreme, not a routine occurrence.

5.3 Import-Heavy but NOT a Shortage - 30 November, 16:00

Timestamp: 30 Nov 2027, 16:00 (a dull late-autumn afternoon) Clearing price: 254.20 EUR/MWh (expensive, but well under the 3,000 shortage ceiling)

Supply source	MWh
Demand	93,543
Renewables (wind+solar+hydro - a dull, low-wind afternoon)	14,088
Biogas	5,289
Conventional (very high output)	53,713
Import (63% of the 30,000 MW ceiling - heavily used but not maxed)	18,910
Storage discharge	1,784

This hour shows import being used heavily as intended - as a genuine relief valve - without the model running out of options entirely. Because import still had roughly 11,000 MW of headroom left below its ceiling, the price rose to reflect real scarcity (254 EUR/MWh, driven mostly by expensive gas/oil plants being needed) without spiking all the way to the 3,000 EUR/MWh shortage ceiling.

5.4 Zero Import, Negative Price - Summer Midday Solar Surplus - 10 June, 12:00

Timestamp: 10 Jun 2027, 12:00 (a sunny midday, solar near its peak) Clearing price: -37.43 EUR/MWh (negative)

Supply source	MWh
Demand	81,485
Renewables (mostly solar - close to peak midday output)	84,858
Biogas	4,778
Conventional (bare minimum still running)	4,837
Import	0 (completely unnecessary)
Storage/flex charging heavily (absorbing the surplus)	12,988

At midday in June, solar output alone (84,858 MWh) already exceeded total demand (81,485 MWh) before even counting biogas or the small amount of conventional generation still running. Storage and the flexible electrolysis/e-mobility agents absorbed nearly 13,000 MWh of the surplus, but that still wasn't enough to prevent the price from going negative - the market had to pay generators to accept a lower price in order to attract more of that absorbing demand. This is the mechanism working exactly as designed: import was correctly not used at all (there was already too much power, importing more would make no sense), and storage did its job of dampening (though not eliminating) the oversupply.

5.5 The Most Negative Price of the Whole Year - 1 September, 08:00

Timestamp: 1 Sep 2027, 08:00 (a very windy, sunny early-autumn morning) Clearing price: -66.46 EUR/MWh (the year's single lowest price)

Supply source	MWh
Demand	81,112
Renewables (wind onshore/offshore + solar + hydro - very high combined output)	78,423
Biogas	4,193
Conventional (only the bare minimum still running)	9,484
Import	0
Storage/flex charging (absorbing as much surplus as their buffers allowed)	10,988

The single lowest price of the year came not from a solar surplus alone but from wind and solar together being unusually strong on the same morning - renewables plus biogas together (82,616 MWh) already exceeded demand, and even close to 11,000 MWh of storage charging couldn't fully absorb the surplus, so price fell further

than on the June solar example above.

Part 6 - The Year in Numbers

6.1 Negative Prices Across the Year

1,608 hours of the real 2027 build (18.4% of the entire year) cleared at a negative price. None occurred January through March. They are concentrated April through November, peaking in May and June - the months where solar and wind output combined is strongest relative to demand (which is itself lower than in winter, since heating load drops off).

Month	Negative-price hours
April	122
May	264
June	270
July	237
August	191
September	189
October	137
November	153
December	45
January - March	0

6.2 Import Usage Across the Year

Import is used as a genuine relief valve, not a routine crutch: it sits at zero for 70% of the year, is used at a partial level for 25%, and is only pushed all the way to its 30,000 MW ceiling for 408 hours (4.7% of the year) - mostly winter hours with low renewable output and high demand, like the two examples in Part 5.2 and 5.3.

Import level	Hours	Share of year
Zero (0 MWh)	6,137	70.0%
Partial (some import, below ceiling)	2,215	25.3%
At ceiling (30,000 MWh)	408	4.7%

Across the full year, total imported energy was 37.2 TWh against 623.5 TWh of total demand - imports covered about 6.0% of Germany's total electricity demand in this simulated year. Mean price across the whole year was 56.18 EUR/MWh, median 70.10 EUR/MWh (the mean sits below the median because negative-price hours pull it down, while the "typical" hour, represented by the median, is comfortably positive).

Part 7 - Extending to 2028 and 2029

2028 and 2029 are out-of-sample validation builds: the same model, the same rules, the same agent structure as 2027, just re-run with each year's own real capacities, demand, and weather-driven profiles. Nothing about HOW the model works (Parts 1-4 above) changes between years - only the numbers that go into it.

7.1 What Stays Identical Across All Three Years

- Every agent type and every rule described in Parts 1-4: the same auction mechanism, the same three-layer conventional generation structure, the same GenericFlexibilityTrader pattern for storage/electrolysis/e-mobility.
- The 30,000 MW import ceiling - kept as the default for 2028 and 2029 too, after a full sweep test (15,000/20,000/30,000 MW) found smaller ceilings improve one narrow metric (excl-shortage correlation) but make the fuller picture (all-hours bias, shortage-hour count) meaningfully worse. Documented as a real, considered trade-off, not an oversight.
- ValueOfLostLoad = 3,000 EUR/MWh, the same shortage-price ceiling.
- The same subsidy structure (FIT/MPVAR rates) and the same price-responsive electrolysis/e-mobility design (Part 3.4) - only the target annual energy volumes and buffer sizes scale up with each year's own Brainpool figures.

7.2 What Changes: Capacity Growth Year to Year

Every capacity figure below is real, taken directly from each year's own agent configuration files - not projected or interpolated. The clear trend: conventional fossil capacity shrinks, while renewables, batteries, electrolysis and e-mobility all grow.

Technology	2027	2028	2029
Lignite	13,927 MW	12,202 MW	7,076 MW
Hard Coal	5,504 MW	4,083 MW	3,732 MW
Natural Gas (combined)	36,237 MW	38,147 MW	40,047 MW
Oil + Other Fossil	9,790 MW	9,578 MW	9,362 MW
Solar Openfield	119,179 MW	132,887 MW	146,121 MW
Solar Rooftop	33,706 MW	40,225 MW	46,736 MW
Wind Onshore	90,232 MW	98,491 MW	106,825 MW
Wind Offshore	12,573 MW	14,994 MW	17,336 MW
Run-of-River	4,161 MW	4,193 MW	4,230 MW
Biogas	8,250 MW	8,250 MW	8,375 MW
Pumped Hydro (fixed asset)	8,378 MW / 53,608 MWh	unchanged	unchanged
Battery Storage	5,152 MW / 10,303 MWh	6,717 MW / 13,433 MWh	9,709 MW / 19,418 MWh
Reservoir Hydro (fixed asset)	1,540 MW / 171,282 MWh	unchanged	unchanged
Electrolysis buffer	1,708 MW avg draw / 100,000 MWh	2,253 MW avg draw / 131,855 MWh	2,834 MW avg draw / 165,889 MWh
E-Mobility buffer	2,033 MW avg draw / 40,000 MWh	2,442 MW avg draw / 48,055 MWh	2,933 MW avg draw / 100,000 MWh
Total annual demand	623.5 TWh	649.7 TWh	676.2 TWh

In plain terms: Battery storage roughly doubles from 2027 to 2029 (10,303 to 19,418 MWh), and both electrolysis and e-mobility grow their mandatory annual draw by 40-65% - all real Brainpool-supplied growth targets, not assumptions we introduced. Fossil capacity, by contrast, keeps shrinking every year - lignite alone loses almost half its 2027 capacity by 2029 (13,927 to 7,076 MW).

7.3 The Leap-Year Handling Difference

2027 and 2029 are both real (non-leap) 365-day years, so both needed the Feb-29-drop calendar bug fix described earlier (dropping Dec 31 instead of Feb 29 from the 2016 demand-shape source, to avoid a mid-year weekday-alignment gap) - see Germany2027_Feb29DropFix and Germany2029_Feb29DropFix. 2028 is a REAL

calendar leap year, but AMIRIS's underlying FAME framework cannot represent a true 366-day year at all (confirmed directly by FAME's own error message: 'last day of leap year is Dec 30th!'). Every AMIRIS year, including 2028, runs as exactly 365 days - FAME itself drops Dec 31 for leap years, not Feb 29. This happens to sidestep the calendar bug automatically for 2028: truncating the (also leap-year) 2016 demand source to its first 8,760 hours drops real Dec-31-2016 and keeps real Feb-29-2016 intact, so no separate fix build was needed for 2028 - the plain Germany2028 build is already correct.

Part 8 - Real Hours From the 2028 and 2029 Results

Two verified example hours per year - a shortage hour and a negative-price hour - mirroring the style of Part 5, pulled directly from each year's real result files and balance-checked the same way.

8.1 2028 Shortage Hour - 18 December, 18:00

Timestamp: 18 Dec 2028, 18:00 (winter evening peak)
ValueOfLostLoad ceiling)

Clearing price: 3,000.00 EUR/MWh (the

Supply source	MWh
Demand	99,749
Renewables (wind+solar+hydro - low, dark winter evening)	12,359
Biogas	5,366
Conventional (running very hard)	46,780
Import (maxed at the 30,000 MW ceiling)	30,000
Storage discharge (drawing down reserves)	8,442
Storage/flex still charging a little elsewhere	3,198

The only shortage hour in the entire real 2028 build - the same pattern as 2027's shortage hour (Part 5.2): a high winter-evening demand peak, weak renewables, conventional generation and import both pushed to their limits, and storage draining reserves, still wasn't quite enough to avoid hitting the price ceiling.

8.2 2028 Most Negative Price - 1 October, 09:00

Timestamp: 1 Oct 2028, 09:00 Clearing price: -77.36 EUR/MWh (the year's single lowest price)

Supply source	MWh
Demand	69,307
Renewables (wind+solar+hydro - strong autumn morning)	77,393
Biogas	4,484
Conventional (minimal)	10,827
Import	0
Storage/flex charging heavily (absorbing the surplus)	23,398

Renewables alone (77,393 MWh) already exceeded total demand (69,307 MWh) that morning. Storage and the flexible electrolysis/e-mobility agents absorbed a very large 23,398 MWh of the surplus - more than double the equivalent 2027 example (Part 5.4/5.5) - reflecting 2028's larger battery and buffer capacities (Part 7.2), yet the oversupply was still large enough to push price to the year's lowest point.

8.3 2029 Shortage Hour - 15 January, 17:00

Timestamp: 15 Jan 2029, 17:00 (winter evening peak)
hours in the real 2029 build)

Clearing price: 3,000.00 EUR/MWh (one of 9 shortage

Supply source	MWh
Demand	103,292
Renewables (wind+solar+hydro - weak)	12,559
Biogas	5,178
Conventional (running very hard)	50,520
Import (maxed at the 30,000 MW ceiling)	30,000
Storage discharge	5,814
Storage/flex still charging a little elsewhere	779

2029 has more shortage hours than 2027 or 2028 (9, vs. 2 and 1 respectively) - directly because 2029's demand

is the highest of the three years (676.2 TWh annual total, Part 7.2) while the import ceiling stays fixed at 30,000 MW. This is exactly the trade-off documented in Germany2029's scenario metadata: growing demand against a fixed import ceiling naturally produces more (rare) shortage hours in later years, even though the ceiling itself was deliberately kept unchanged after the full sweep evaluation (Part 7.1).

8.4 2029 Most Negative Price - 25 November, 09:00

Timestamp: 25 Nov 2029, 09:00 Clearing price: -85.10 EUR/MWh (the lowest price of any year built so far)

Supply source	MWh
Demand	74,101
Renewables (wind+solar+hydro - very strong for late November)	82,341
Biogas	5,328
Conventional (minimal)	7,275
Import	0
Storage/flex charging heavily	20,843

The single most negative price across all three built years. As in the other examples, renewables alone exceeded demand, and despite over 20,000 MWh of storage/flex charging absorbing much of the surplus, the market still had to clear well below zero.

Part 9 - Cross-Year Summary: How the Model Performs Out-of-Sample

2027 is the calibrated year (Brainpool's real 2027 data was used to build and check the model in the first place). 2028 and 2029 are genuine out-of-sample tests - the same model structure, re-run on each year's own real inputs, then compared against Brainpool's own real forecast for that year, which the model never saw during calibration.

Metric	2027	2028	2029
Correlation vs. Brainpool (all hours)	$r = 0.515$	$r = 0.394$	$r = 0.475$
AMIRIS mean price	56.18 EUR/MWh	50.79 EUR/MWh	56.35 EUR/MWh
Brainpool real mean price	68.04 EUR/MWh	64.02 EUR/MWh	61.85 EUR/MWh
AMIRIS shortage hours (price = 3,000)	2	1	1
Brainpool's own extreme hours ($\geq 3,000$ EUR/MWh)	0	1 (4,000 EUR/MWh)	3 (4,000 EUR/MWh)
Negative-price hours	1,608 (18.4%)	1,793 (20.5%)	1,780 (20.3%)
Import used at ceiling	408 hrs (4.7%)	389 hrs (4.4%)	403 hrs (4.6%)
Import used at all	30.0% of hours	26.9% of hours	31.5% of hours

The correlation drop from 2027 (0.515) to 2028/2029 (0.394 / 0.475) is the expected, honest signature of out-of-sample testing - the model was never tuned against 2028 or 2029 data, only built from each year's own real inputs the same mechanical way 2027 was. Both out-of-sample years still track Brainpool's real price meaningfully (correlation well above zero, and mean prices within about 10-25% of Brainpool's own), while negative-price and import-usage patterns stay close to 2027's in relative terms - evidence that the model's core mechanics (Parts 1-4) generalise across years, even though absolute price-matching accuracy naturally weakens somewhat outside the calibrated year.